

Multi-Model Boundary Refinement for Person Segmentation

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Abstract

Person segmentation is useful in image editing, human-centered vision, and background removal tasks, but final mask boundaries are often unstable around hair, clothing, shadows, and complex backgrounds. This project studies a practical multi-model pipeline for improving person boundary segmentation without training a new neural network. The workflow combines YOLO-based person localization, SAM-based mask generation, Canny edge detection, edge-based morphological refinement, and GrabCut foreground refinement.

Method Summary

Baseline v1: automatic SAM mask generation with Canny edge scoring and GrabCut.
Proposed v6: YOLO person detection creates an automatic SAM box prompt. The resulting mask is refined by Canny-guided edge refinement and GrabCut.

Experimental Setup

The formal test set contains 12 selected person images from 34 uploaded photos. Six automatic versions were batch-tested. Manual box or point prompt versions were prepared but skipped in the formal batch because manual prompt annotations were not included.

Main Quantitative Result

v1 SAM auto + Canny score + GrabCut: mean edge alignment = 0.8708.
v6 YOLO detection + SAM + Canny + edge refine + GrabCut: mean edge alignment = 0.8855.
Absolute improvement over v1: 0.0147. Relative improvement: about 1.7 percent.

Interpretation

The result supports a modest preliminary improvement in edge alignment. YOLO helps reduce target ambiguity by providing an automatic person box prompt for SAM. Edge refinement and GrabCut improve boundary stability in the selected test set.

Strict Limitations

This project does not report IoU or Dice score because manually annotated ground-truth masks are not included. Edge alignment is a proxy metric and should not be interpreted as complete segmentation accuracy. The dataset is small, so the result should not be claimed as a universal improvement over SAM, YOLO-Seg, or supervised segmentation models.

References Used

- [1] Segment Anything, ICCV 2023.
- [2] A Comprehensive Review of YOLO Architectures in Computer Vision, 2023.
- [3] SAM prompt evaluation studies.
- [4] Canny edge detection.
- [5] GrabCut foreground extraction.